

ST LEONARD, NB, Canada

WMO: 717030

Lat: 47.158N

Lon: 67.832W

Elev: 795

StdP: 14.28

Time Zone: -4.00 (NAA)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-10.8	-22.1	1.7	-14.9	-17.4	2.2	-10.2	27.3	12.7	24.1	13.2	5.9	300	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	83.7	68.2	80.1	66.0	77.2	64.6	71.0	78.7	69.2	76.1	67.4	73.6	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.5	107.8	74.6	66.8	101.6	72.8	65.1	95.7	70.9	35.4	78.3	33.9	76.4	32.3	73.5	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.2	16.7	-23.9	88.7	5.3	2.3	-27.7	90.3	-30.8	91.7	-33.8	93.0	-37.7	94.7		
(5)			-23.9	74.1	5.4	2.1	-27.9	75.6	-31.0	76.9	-34.1	78.0	-38.0	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	10.4	13.2	23.9	37.0	50.7	60.0	65.2	63.1	55.2	43.2	31.6	18.7		
	DBStd	20.92	12.72	11.73	10.64	7.67	7.70	7.05	5.24	5.85	7.40	7.73	9.24	11.05		
	HDD50	5349	1227	1030	810	395	87	5	0	1	30	239	555	970		
	HDD65	9477	1692	1450	1274	841	448	184	62	106	306	676	1003	1435		
	CDD50	1511	0	0	1	4	107	304	472	408	185	28	2	0		
	CDD65	163	0	0	0	0	3	33	68	48	11	0	0	0		
	CDH74	1438	0	0	0	2	93	321	524	403	92	3	0	0		
	CDH80	297	0	0	0	0	18	73	99	91	16	0	0	0		
	WSAvg	7.0	7.7	8.0	8.1	7.8	7.2	6.5	5.7	5.1	5.9	7.0	7.4	7.5		
	PrecAvg	41.7	3.0	2.5	2.8	2.9	3.5	4.0	4.3	3.9	3.9	4.1	3.3	3.5		
(15) Precipitation	PrecMax	52.2	5.3	4.2	6.1	5.2	5.7	7.0	7.6	8.8	7.5	9.2	7.7	6.7		
	PrecMin	29.6	0.6	0.7	1.1	1.3	1.7	1.6	1.3	1.9	1.7	1.1	0.7	1.4		
	PrecStd	5.1	1.4	0.7	1.3	1.1	1.1	1.3	1.5	1.7	1.5	1.9	1.6	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	44.0	52.9	67.4	82.3	86.4	86.9	86.7	82.0	71.5	60.1	46.4		
		MCWB	44.1	41.3	43.3	52.4	61.0	69.2	69.9	69.5	69.9	59.3	57.6	45.3		
	2%	DB	38.6	38.3	45.6	59.3	76.1	81.9	82.7	82.6	76.1	63.9	53.5	40.5		
		MCWB	37.5	35.9	38.9	47.7	57.8	65.9	68.8	68.1	65.7	55.8	49.7	39.1		
	5%	DB	33.7	34.1	41.3	54.1	70.4	77.6	79.5	78.3	71.8	59.4	48.4	36.8		
		MCWB	32.4	32.3	36.3	44.6	55.6	63.7	66.5	65.8	63.8	54.1	45.5	35.2		
	10%	DB	29.1	31.0	38.1	49.5	65.6	73.7	76.6	75.0	68.2	55.7	44.6	33.8		
		MCWB	27.9	29.4	34.1	41.6	53.8	62.2	65.2	64.4	61.4	51.0	42.2	32.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.5	42.2	47.1	55.9	65.7	72.2	74.2	72.4	71.5	62.0	57.9	45.8		
		MCDB	44.9	44.0	49.7	63.3	74.1	82.0	81.4	82.1	80.3	66.7	59.4	46.4		
	2%	WB	37.6	35.9	40.4	50.0	61.5	69.2	71.3	70.2	68.2	58.0	51.0	39.4		
		MCDB	38.4	37.4	43.3	55.5	69.1	77.7	78.0	77.4	73.2	62.1	52.6	40.2		
	5%	WB	32.4	32.5	37.0	45.8	58.4	66.8	69.2	68.3	65.4	55.0	46.0	35.5		
		MCDB	33.6	34.1	40.7	52.5	66.3	73.8	76.0	74.8	69.6	59.2	47.7	36.4		
	10%	WB	28.1	29.4	34.3	42.4	55.4	64.4	67.3	66.5	62.6	51.5	42.4	32.7		
		MCDB	29.1	31.0	37.7	48.5	64.0	70.3	73.4	72.5	67.1	55.1	44.5	33.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	18.0	17.5	17.3	20.6	20.5	19.2	20.3	20.0	16.1	12.8	14.1		
		MCDBR	20.4	17.1	19.6	26.4	30.2	26.8	23.9	25.1	23.9	23.0	17.8	14.8		
		MCWBR	20.4	15.5	14.8	17.1	15.9	14.1	12.2	13.1	14.5	16.1	15.7	14.3		
		MCDBR	20.0	15.7	17.0	22.6	22.9	21.2	19.5	19.7	19.1	20.1	16.1	14.5		
	5% WB	MCWBR	20.1	14.8	14.0	16.8	14.4	12.8	11.3	11.5	13.2	15.7	15.5	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.270	0.301	0.341	0.351	0.370	0.403	0.382	0.356	0.318	0.289	0.272		
	taud		2.345	2.283	2.278	2.368	2.422	2.369	2.297	2.357	2.422	2.526	2.527	2.377		
	Ebn at Noon		269	286	292	290	290	283	272	273	271	267	256	246		
	Edh at Noon		23	30	35	36	35	37	40	36	31	24	19	20		
(44) All-Sky Solar Radiation	RadAvg		439	720	1098	1410	1590	1724	1718	1505	1180	705	397	305		
	RadStd		54	83	76	127	160	121	148	94	90	64	48	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page